# CHEN Chaofeng (陈超锋)

**Tenure Track Assistant Professor** | School of Artificial Intelligence, Wuhan University

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## RESEARCH INTERESTS

Computer vision and image processing, with a focus on image/video quality assessment, image restoration and enhancement, and multi-modality generative models.

## RESEARCH EXPERIENCE

**Tenure Track Assistant Professor** Present
School of Artificial Intelligence, Wuhan University

**Postdoctoral Research Fellow** Sep. 2021 - Jan. 2025
S-Lab, Nanyang Technological University; worked with Prof. Weisi Lin and Prof. Kelly

## EDUCATION

**The University of Hong Kong (HKU)** Sep. 2015 - Jan. 2021
Ph.D. in Computer Science | Supervisor: Prof. Kenneth K. Y. Wong

**Huazhong University of Science and Technology (HUST)** Sep. 2011 - Jun. 2015
B.Eng. in Computer Science

## SELECTED PUBLICATIONS

Full list: Google Scholar. * Equal contribution; # Corresponding author.

### Recent Publications (2025-2026)

[1] **[MM 2026]** GeoMAR: Unleashing Geometrically Aligned Features for Masked Autoregressive Blind Face Restoration. Lu Gan, Hanyu Yan, **Chaofeng Chen**, Junqi Hu, Dan Zeng.

[2] **[MM 2026]** Decoupled Similarity for Task-Aware Token Pruning in Large Vision-Language Models. Kexin Ma, Jing Xiao, **Chaofeng Chen**, Geyong Min, Guibo Zhu, Jinqiao Wang, Liang Liao.

[3] **[MM 2026; Oral]** ETPDesigner: Multi-Agent Orchestration for Interactive Multimodal Electronic Theater Program. Mengtian Li, Xinru Guo, Xiaoru Lin, Xiao Rong, Zhifeng Xie, **Chaofeng Chen**#.

[4] **[TPAMI 2026]** Text4Seg++: Advancing Image Segmentation via Generative Language Modeling. Mengcheng Lan, **Chaofeng Chen**, Jiaxing Xu, Zongrui Li, Yiping Ke, Xudong Jiang, Yingchen Yu, Yunqing Zhao, Song Bai.

[5] **[TMM 2026]** Prompt-Image-Caption Consistency for AI-Generated Image Quality Assessment. Wen Sun, **Chaofeng Chen**, Liang Liao, Weisi Lin.

[6] **[TMM 2025]** Denoised Semantic Features for Local Consistent No-Reference Image Quality Assessment. Hui Li, Liang Liao, **Chaofeng Chen**, Xiaopeng Fan, Wangmeng Zuo, Weisi Lin.

[7] **[ICCV 2025; Highlight]** MVQA: Mamba with Unified Sampling for Efficient Video Quality Assessment. Yachun Mi, Yu Li, Weicheng Meng, **Chaofeng Chen**, Chen Hui, Shaohui Liu.

[8] **[ICLR 2025]** Text4Seg: Reimagining Image Segmentation as Text Generation. Mengcheng Lan, **Chaofeng Chen**, Yue Zhou, Jiaxing Xu, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang.

### Selected Conference Papers

[9] **[MM 2024; Oral]** Q-Ground: Image Quality Grounding with Large Multi-modality Models. **Chaofeng Chen**, Yang Sensen, Haoning Wu, Liang Liao, Zicheng Zhang, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin.

[10] **[ECCV 2024]** Enhancing Diffusion Models with Text-Encoder Reinforcement Learning. **Chaofeng Chen***, Annan Wang*, Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, Weisi Lin#.

[11] **[AAAI 2024]** Iterative Token Evaluation and Refinement for Real-World Super-Resolution. **Chaofeng Chen**, Shangchen Zhou, Liang Liao, Haoning Wu, Wenxiu Sun, Qiong Yan, Weisi Lin.

[12] **[ICML 2024]** Q-Align: Teaching LMMs for Visual Scoring via Discrete Text-Defined Levels. Haoning Wu[*], Zicheng Zhang[*], Weixia Zhang, **Chaofeng Chen**, Liang Liao, Chunyi Li, Yixuan Gao, Annan Wang, Erli Zhang, Wenxiu Sun, Qiong Yan, Xiongkuo Min, Guangtao Zhai, Weisi Lin[#].

[13] **[ECCV 2024]** ClearCLIP: Decomposing CLIP Representations for Dense Vision-Language Inference. Mengcheng Lan, **Chaofeng Chen**, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang.

[14] **[ECCV 2024]** ProxyCLIP: Proxy Attention Improves CLIP for Open-Vocabulary Segmentation. Mengcheng Lan, **Chaofeng Chen**, Yiping Ke, Xinjiang Wang, Litong Feng, Wayne Zhang.

[15] **[CVPR 2024]** Local Distortion Aware Efficient Transformer Adaptation for Image Quality Assessment. Kangmin Xu, Liang Liao, Jing Xiao, **Chaofeng Chen**, Haoning Wu, Qiong Yan, Weisi Lin.

[16] **[CVPR 2024]** Q-Instruct: Improving Low-level Visual Abilities for Multi-modality Foundation Models. Haoning Wu[*], Zicheng Zhang[*], Erli Zhang[*], **Chaofeng Chen**, Liang Liao, Annan Wang, Kaixin Xu, Chunyi Li, Jingwen Hou, Guangtao Zhai, Geng Xue, Wenxiu Sun, Qiong Yan, Weisi Lin[#].

[17] **[ICLR 2024; Spotlight]** Q-Bench: A Benchmark for General-Purpose Foundation Models on Low-level Vision. Haoning Wu[*], Zicheng Zhang[*], Erli Zhang[*], **Chaofeng Chen**, Liang Liao, Annan Wang, Chunyi Li, Wenxiu Sun, Qiong Yan, Guangtao Zhai, Weisi Lin[#].

[18] **[ICCV 2023]** Exploring Video Quality Assessment on User Generated Contents from Aesthetic and Technical Perspectives. Haoning Wu, Erli Zhang, Liang Liao, **Chaofeng Chen**, Jingwen Hou, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin.

[19] **[AAAI 2023]** MIMO Is All You Need: A Strong Multi-In-Multi-Out Baseline for Video Prediction. Shuliang Ning, Mengcheng Lan, Yanran Li, **Chaofeng Chen**, Qian Chen, Xunlai Chen, Xiaoguang Han, Shuguang Cui.

[20] **[MM 2022; Oral]** Real-World Blind Super-Resolution via Feature Matching with Implicit High-Resolution Priors. **Chaofeng Chen**[*], Xinyu Shi[*], Yipeng Qin, Xiaoming Li, Xiaoguang Han, Tao Yang, Shihui Guo.

[21] **[NeurIPS 2022]** S3-NeRF: Neural Reflectance Field from Shading and Shadow under a Single Viewpoint. Wenqi Yang, Guanying Chen, **Chaofeng Chen**, Zhenfang Chen, Kwan-Yee K. Wong.

[22] **[ECCV 2022]** FAST-VQA: Efficient End-to-end Video Quality Assessment with Fragment Sampling. Haoning Wu, **Chaofeng Chen**, Jingwen Hou, Liang Liao, Annan Wang, Wenxiu Sun, Qiong Yan, Weisi Lin.

[23] **[ECCV 2022]** PS-NeRF: Neural Inverse Rendering for Multi-view Photometric Stereo. Wenqi Yang, Guanying Chen, **Chaofeng Chen**, Zhenfang Chen, Kwan-Yee K. Wong.

[24] **[ECCV 2022]** From Face to Natural Image: Learning Real Degradation for Blind Image Super-Resolution. Xiaoming Li, **Chaofeng Chen**, Xianhui Lin, Wangmeng Zuo, Lei Zhang.

[25] **[MM 2022; Oral]** Exploring the Effectiveness of Video Perceptual Representation in Blind Video Quality Assessment. Liang Liao, Kangmin Xu, Haoning Wu, **Chaofeng Chen**, Wenxiu Sun, Qiong Yan, Weisi Lin.

[26] **[CVPR 2021]** Progressive Semantic-Aware Style Transformation for Blind Face Restoration. **Chaofeng Chen**, Xiaoming Li, Lingbo Yang, Xianhui Lin, Lei Zhang, Kwan-Yee K. Wong.

[27] **[ICCV 2021]** HDR Video Reconstruction: A Coarse-to-fine Network and A Real-world Benchmark Dataset. Guanying Chen, **Chaofeng Chen**, Shi Guo, Zhetong Liang, Kwan-Yee K. Wong, Lei Zhang.

[28] **[ECCV 2020]** Blind Face Restoration via Deep Multi-scale Component Dictionaries. Xiaoming Li, **Chaofeng Chen**, Shangchen Zhou, Xianhui Lin, Wangmeng Zuo, Lei Zhang.

### Selected Journal Papers

[29] **[TIP 2024; ESI Highly Cited Paper]** TOPIQ: A Top-down Approach from Semantics to Distortions for Image Quality Assessment. **Chaofeng Chen**, Jiadi Mo, Jingwen Hou, Haoning Wu, Liang Liao, Wenxiu Sun, Qiong Yan, Weisi Lin.

[30] **[CVIU 2023]** Semi-supervised Cycle-GAN for face photo-sketch translation in the wild. **Chaofeng Chen**, Wei Liu, Xiao Tan, Kwan-Yee K. Wong.

[31] **[TPAMI 2023]** Neighbourhood Representative Sampling for Efficient End-to-end Video Quality Assessment. Haoning Wu, **Chaofeng Chen**, Liang Liao, Jingwen Hou, Wenxiu Sun, Qiong Yan, Jinwei Gu, Weisi Lin.

[32] **[TCSVT 2023]** DisCoVQA: Temporal Distortion-Content Transformers for Video Quality Assessment. Haoning Wu, **Chaofeng Chen**, Liang Liao, Jingwen Hou, Wenxiu Sun, Qiong Yan, Weisi Lin.

[33] **[TIP 2023]** Deep Face Video Inpainting via UV Mapping. Wenqi Yang, Zhenfang Chen, **Chaofeng Chen**, Guanying Chen, Kwan-Yee K. Wong.

[34] **[TIP 2021]** Learning Spatial Attention for Face Super-Resolution. **Chaofeng Chen**, Dihong Gong, Hao Wang, Zhifeng Li, Kwan-Yee K. Wong.

### Preprints

[35] **[arXiv preprint 2025]** GaussianMorphing: Mesh-Guided 3D Gaussians for Semantic-Aware Object Morphing. Mengtian Li, Yunshu Bai, Yimin Chu, Yijun Shen, Zhongmei Li, Weifeng Ge, Zhifeng Xie, **Chaofeng Chen**[#].

[36] **[arXiv preprint 2025]** 2D Gaussian Splatting with Semantic Alignment for Image Inpainting. Hongyu Li, **Chaofeng Chen**, Xiaoming Li, Guangming Lu.

## ACADEMIC ACTIVITIES

**Area Chair:** ICLR 2027; ICML 2027

**Senior Program Committee Member:** AAAI 2027

**Forums:**

- PRCV 2026: Invited forum talk on quality assessment and reward modeling for generative vision (Aug. 2026).
- CCIG 2026: Forum Chair, Low-Quality Visual Processing and Quality Assessment (May 2026).

**Committee Member:**

- Member, Multimedia Technical Committee, China Society of Image and Graphics (CSIG).
- Executive Member, Multimedia Technical Committee, China Computer Federation (CCF).

**Conference Reviewer:** CVPR, ICCV, ECCV, SIGGRAPH Asia, ACM MM, ICLR, NeurIPS, ICML, AAAI.

**Journal Reviewer:** TPAMI, IJCV, TIP, CVIU, TIFS, TNNLS, Neurocomputing, TMM, TCSVT.

## TEACHING

| Institution / Term | Course | Role |
|---|---|---|
| **WHU**<br>Spring 2026 | Introduction to Machine Learning | Lecturer |
| **SYSU**<br>Spring 2025 | AI236 Principles of Artificial Intelligence<br>AI238 Experiment of Artificial Intelligence | Lecturer |

## HONORS AND AWARDS

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|---|---|
| World's Top 2% Scientists, Stanford University | 2025, 2026 |
| ICML Silver Reviewer | 2026 |
| Hong Kong PhD Fellowship | 2015 - 2018 |
| National Scholarship | 2013 - 2014 |